US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12410069
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Shrader; Martin et al.

Wastewater processing

Abstract

A wastewater processing method includes introducing wastewater into an upper region of a chamber. The chamber remains at substantially atmospheric pressure. A portion of the wastewater in the chamber is vaporized. Flame is introduced into the chamber and provides for the ignition of a volatile organic compound. The vaporized portion of the wastewater is vented to the atmosphere.

Inventors: **Shrader; Martin (Indianapolis, IN), Reece, Jr.; Merrill (Indianapolis, IN), Haag; Anthony (Brownsburg, IN), Kite; Paul (Zionsville, IN)**

Applicant: **M2 Water Solutions LLC** (Indianapolis, IN)

Family ID: **84977053**

Assignee: **M2 Water Solutions, LLC (Indianapolis, IN)**

Appl. No.: **17/956209**

Filed: **September 29, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230028512 A1	Jan. 26, 2023

Related U.S. Application Data

continuation-in-part parent-doc US 17093511 20201109 US 11479479 child-doc US 17956209
continuation-in-part parent-doc US 17069005 20201013 US 11401173 20220802 child-doc US 17093511
division parent-doc US 16192083 20181115 US 10807884 20201020 child-doc US 17069005
us-provisional-application US 62933416 20191109
us-provisional-application US 62933414 20191109

Publication Classification

Int. Cl.: **C02F1/04** (20230101); **B01D1/00** (20060101); **B01D1/16** (20060101)

U.S. Cl.:

CPC **C02F1/048** (20130101); **B01D1/0058** (20130101); **B01D1/0082** (20130101); **B01D1/16** (20130101);

Field of Classification Search

CPC: C02F (1/048); B01D (1/0058); B01D (1/0082); B01D (1/16)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3212559	12/1964	Williamson	159/13.4	B01D 3/06
4045314	12/1976	Rod et al.	N/A	N/A
4838184	12/1988	Young	N/A	N/A
5183577	12/1992	Lehmann	N/A	N/A
5207869	12/1992	Harmoning	159/901	B01D 1/0082
5256251	12/1992	Holcombe	N/A	N/A
5934207	12/1998	Echols	110/345	F23G 7/008
6623645	12/2002	Road et al.	N/A	N/A
6887344	12/2004	VanKouwenberg	N/A	N/A
7228682	12/2006	Kashler	N/A	N/A
7448600	12/2007	Boulter	N/A	N/A
7604710	12/2008	Haslem et al.	N/A	N/A
7681643	12/2009	Heins	N/A	N/A
7722739	12/2009	Haslem et al.	N/A	N/A
8211303	12/2011	Page et al.	N/A	N/A
8216468	12/2011	Page et al.	N/A	N/A
8256748	12/2011	Boulter	N/A	N/A
8282772	12/2011	Jepson	N/A	N/A
8343310	12/2012	Fielding et al.	N/A	N/A
8425668	12/2012	Lakatos et al.	N/A	N/A
8470139	12/2012	Booth	N/A	N/A
8679291	12/2013	Duesel, Jr. et al.	N/A	N/A
8764947	12/2013	Tang et al.	N/A	N/A
8877690	12/2013	Keister	N/A	N/A
8906203	12/2013	Ugolin	N/A	N/A
9446974	12/2015	Milner et al.	N/A	N/A
9580343	12/2016	Keister	N/A	N/A
9770671	12/2016	Curlett	N/A	N/A
10092851	12/2017	Dighe	N/A	N/A
10807884	12/2019	Shrader	N/A	B01D 1/0082
11401173	12/2021	Shrader	N/A	B01D 1/0058
11479479	12/2021	Shrader	N/A	B01D 1/16

2011/0140457	12/2010	Lakatos et al.	N/A	N/A
2012/0247942	12/2011	Curlett	N/A	N/A
2016/0045841	12/2015	Kaplan	429/49	C01B 32/05
2017/0282084	12/2016	Wan et al.	N/A	N/A
2017/0282094	12/2016	Wan	N/A	N/A

OTHER PUBLICATIONS

Webpage for Colaris Thermal Process Technology. https://caloris.com/caloris-agilix-mobile-evaporator/. Accessed Feb. 18, 2019 online. cited by applicant

Primary Examiner: Miller; Jonathan

Attorney, Agent or Firm: Indiano Law Group LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. application Ser. No. 17/093,511, filed Nov. 9, 2020, and entitled “Wastewater Processing”, which is a continuation-in-part of U.S. application Ser. No. 17/069,005 (filed Oct. 13, 2020, and having issued as U.S. Pat. No. 11,401,173), which is a divisional of U.S. application Ser. No. 16/192,083 (filed Nov. 15, 2018 and having issued as U.S. Pat. No. 10,807,884), claims the benefit of U.S. Provisional Application No. 62/933,416 (filed Nov. 9, 2019, and entitled “Quick Change Nozzle System for Water Processing System and Method for Quickly Changing and Maintaining Nozzles in a Process Spray System”), and claims the benefit of U.S. Provisional Application No. 62/933,414 (filed Nov. 9, 2019, and entitled “Remotely Monitored and Controlled Wastewater Processing at Atmospheric Pressure and Method for Remotely Monitoring and Controlling a Wastewater Processing System”), the entireties of which are incorporated by reference herein.

TECHNICAL FIELD OF THE INVENTION
(1) The present application relates to wastewater and more particularly, but not exclusively to, wastewater processing.
BACKGROUND
(2) Various agricultural and industrial processes result in the production of wastewater. For example, wastewater is often generated as a byproduct of oil and gas production, textile production, mining and mineral production, etc.
(3) The oil and gas industry utilizes water for a variety of tasks and generates a significant amount of wastewater. For example, water may be injected into a well to re-pressurize a reservoir, water may be pumped from a well in the process of extracting oil or gas, and water may be used to deliver proppants to underground fractures. During such uses, various hydrocarbons (which can include crude oil as well as various fats and other oils), heavy metals, various volatile organic compounds, various other noxious compounds, and various solids are dissolved by and/or introduced into the water. This contaminated water is referred to as “wastewater” typically cannot be directly released back into the environment due to the noxious compounds therein.
(4) Significant operational costs are incurred by oil and gas producers related to the disposal or re-use of such wastewater. Many other high-volume wastewater generators face similar significant operational costs for disposal of wastewater.
(5) Wastewater processing often involves one or more unit operations, such as distillation, evaporation, filtration, and vaporization, to reduce the total volume of wastewater (e.g., by

reducing the amount of water present in the wastewater). Distillation is an energy-intensive process that frequently requires large distillation columns. Filtration may require frequent filter changes to keep the system operating at the desired contaminant levels. Vaporization can require various pre-conditioning and high-pressure operations.
(6) Many wastewater processing units are large, fixed installations which are not suitable for mid-volume locations or remote locations. Such wastewater processing units can be very expensive. Wastewater is often transported from remote locations, often over long distances, for processing and/or disposal. The costs associated with the transportation and disposal of wastewater can add significantly to the production costs of the underlying business producing the wastewater.
(7) Additionally, some evaporative and vaporization wastewater processing units of the prior art have been known to release various noxious compounds into the air (e.g., such as volatile organic compounds) during the evaporative and vaporization processes. Due to the numerous drawbacks of processing and disposal systems of the prior art, further technological developments are desirable in this area.

SUMMARY

(8) One embodiment of the present application includes a wastewater processing method. This method includes introducing wastewater into an upper region of a chamber. The chamber remains at substantially atmospheric pressure. A portion of the wastewater is vaporized in the chamber. Flame is introduced into the chamber. A volatile organic compound is ignited. This ignition can be due to the interaction of the flame with the volatile organic compound. The vaporized portion of the wastewater is vented.
(9) The method can include introducing the flame into a lower region of the chamber. A vaporization medium can be provided in the upper region of the chamber. Introducing wastewater into the upper region of the chamber can include directing wastewater toward the vaporization medium.
(10) The method can include providing a permeable substrate between the upper region and the lower region. The vaporization medium can be a plurality of pall rings.
(11) The method can include vaporizing a substantial portion of the wastewater as the wastewater moves downwardly through the upper region. The flame can be directed into a lower region of the chamber and the flame can contact un-vaporized droplets of the wastewater as the un-vaporized droplets move downwardly through the lower region.
(12) Introducing flame into the chamber can include introducing flame into the chamber at a temperature between 500° F. to 3500° F. The method can include collecting an un-vaporized portion of the wastewater in a lower region of the chamber.
(13) The method can include introducing the flame into the lower region of the chamber. The flame can be directed toward an upper surface of the collected un-vaporized portion of the wastewater so that the flame contacts the upper surface of the collected un-vaporized portion. The lower chamber can be an expansion chamber for the flame (e.g., the lower chamber provides a sufficient volume for the flame to expand, and the lower chamber can provide a sufficient volume for the fuel which is generating the flame to combust). A blower can introduce air into the lower chamber.
(14) The method can include determining a current concentration of a waste within the un-vaporized portion of wastewater. The waste can include at least one of dissolved solids, suspended solids, and volatile organic compounds (e.g., the waste can include dissolved solids, suspended solids, and/or volatile organic compounds). The current concentration can be compared with a predetermined threshold concentration. In response to the current concentration being equal to or greater than the predetermined threshold concentration, the method can include disposing of the collected un-vaporized portion.
(15) In response to the current concentration being lower than the predetermined threshold concentration, the method can include introducing the collected un-vaporized portion into the upper region of the chamber as wastewater (e.g., the collected un-vaporized liquid can be circulated

through the chamber until the amount of waste present within the collected un-vaporized liquid is equal to or greater than the predetermined threshold concentration). In this manner, much of the water present within the wastewater can be removed from the wastewater, and a concentrated liquid waste (e.g., having a significantly reduced volume relative a volume of raw wastewater which was introduced into the chamber) can be disposed of.
(16) Another embodiment of the present application is directed to an apparatus. This apparatus includes a chamber having an outer wall. A wastewater introduction port is located at an upper portion of the chamber. The wastewater introduction port is configured to release wastewater into the upper portion. A burner is placed in flow communication with the chamber, and the burner is configured to generate flame to be introduced into the chamber. A portion of the wastewater is vaporized in the upper chamber.
(17) The flame can be introduced into the lower portion of the chamber. The flame can have a temperature in the range of 500° F. to 3500° F., when the flame is introduced into the lower portion. The flame can interact with at least one volatile organic compound in the chamber to ignite and incinerate the at least one volatile organic compound.
(18) A permeable substrate can be located between the upper portion and the lower portion. The upper portion can be an upper region, the lower portion can be a lower region, and a vaporization medium can be located in the upper region. The wastewater introduction port can be configured to expel the wastewater toward the vaporization medium.
(19) A vent can be placed in flow communication with the upper region of the chamber, and the portion of vaporized wastewater can be vented to the atmosphere via the vent. The chamber can remain at substantially atmospheric pressure.
(20) A burner tube can extend from the burner through the upper region and toward the lower region (e.g., a first end of the burner tube can be flowably coupled with an outlet of the burner and a second end of the burner tube can terminate at the lower region, and the burner tube can extend through the upper region). The upper region is defined between the outer wall of the chamber and an outer wall of the burner tube. The lower region of the chamber can be configured to collect un-vaporized wastewater. The burner tube can be configured to direct the flame downwardly at a surface of the collected un-vaporized wastewater, and the flame can contact the surface of the collected un-vaporized wastewater.
(21) A vaporization medium can be located in the upper portion of the chamber. The flame can be introduced into the lower portion of the chamber and the lower portion can be configured to serve as an expansion chamber for the flame. The fuel generating the flame can be fully combusted and converted into heated exhaust gas in the lower portion. The heated exhaust gas can be directed upwardly toward the vaporization medium and can flow through the vaporization medium.
(22) A substantial portion of the wastewater can be vaporized in the upper portion. Un-vaporized droplets of the wastewater can move downwardly from the upper portion into the lower portion and the flame can contact the un-vaporized droplets as they move downwardly in the lower portion.
(23) Yet another embodiment of the present application is directed to a method that includes introducing wastewater into an upper portion of a chamber and vaporizing a substantial portion of the wastewater. The method includes igniting a volatile compound that was released from the wastewater during the vaporizing. The method includes venting the vaporized portion of the wastewater to the atmosphere.
(24) The volatile compound can be a volatile organic compound. The method can include introducing flame into a lower portion of the chamber, and the flame can be configured to be an ignition source for the volatile organic compound.
(25) The method can include introducing flame into a lower portion of the chamber. The flame can include a temperature selected from the range of 1000° F. to 2500° F., when introduced into the lower portion. The lower portion of the chamber can be an expansion chamber for the flame.
(26) The method can include maintaining the chamber at substantially atmospheric pressure. An

un-vaporized portion of the wastewater can be collected in the lower portion of the chamber. The flame can be directed downwardly toward a surface of the collected un-vaporized wastewater and the flame can contact the surface of the collected un-vaporized wastewater. A fuel which feeds the flame (e.g., the fuel which is combusting and generating the flame) can be combusted and converted into heated exhaust gas in the lower portion of the chamber. The heated exhaust gas can flow upwardly toward the upper portion of the chamber.
(27) A vaporization medium can be provided in the upper portion of the chamber. Introducing wastewater into an upper portion of a chamber can include spraying the wastewater upon the vaporization medium. The method can include flowing the heated exhaust gas upwardly through the vaporization medium. The substantial portion of the wastewater can be vaporized at the vaporization medium (e.g., a substantial portion of the wastewater can be vaporized in, on, or near the vaporization medium as the wastewater moves downwardly through the vaporization medium and the heated exhaust gas flows upwardly through the vaporization medium).
(28) Other embodiments include unique wastewater processing apparatuses, systems, and methods. Further embodiments, inventions, forms, objects, features, advantages, aspects, and benefits of the present application are otherwise set forth or become apparent from the description and drawings included herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The description herein makes reference to the accompanying drawings wherein like reference numerals refer to like parts throughout the several views, and wherein:
(2) FIG. **1** is a schematic block diagram of an exemplary wastewater processing system;
(3) FIG. **2** is a schematic cross-sectional view of an exemplary wastewater collection system, shown with the pall rings removed;
(4) FIG. **3**A is a schematic cross-sectional view of an exemplary vaporization system;
(5) FIG. **3**B is a schematic cross-sectional view of another exemplary vaporization system, with flame introduced into a lower region;
(6) FIGS. **4**A and **4**B are schematic depictions of exemplary water introduction components;
(7) FIG. **4**C is a schematic cutaway top view of the vaporization system of FIG. **3**B, shown with the water introduction components of FIG. **4**A installed;
(8) FIG. **5** depicts an exemplary pall ring;
(9) FIG. **6**. is a schematic plan view of an exemplary water transfer system of the wastewater processing system of FIG. **1**;
(10) FIG. **7** is a schematic cross-sectional view of an exemplary mobile wastewater processing system;
(11) FIG. **8** is an enlarged view of the expansion chamber of the vaporization system of FIG. **3**B;
(12) FIG. **9** is a side elevation view of yet another exemplary vaporization system, with the outer cylinder transparent, shown with the pall rings removed;
(13) FIGS. **10**A and **10**B are side elevation views of an alternative exemplary collection chamber, shown with the vapor chamber and burner tube removed for clarity;
(14) FIG. **11** is a perspective cutaway view of the vaporization system of FIG. **9**, depicting the vapor chamber, shown with the pall rings removed;
(15) FIG. **12** is a perspective view of one spray rail assembly of FIG. **11**, shown removed from the vaporization system;
(16) FIG. **13** is a perspective view of the vaporization system of FIG. **9**, showing a water supply coupled with one of the spray rail assemblies; and
(17) FIG. **14** is an enlarged view of the vaporization system of FIG. **9**, depicting the collection

chamber during use.
(18) The accompanying drawings incorporated in and forming a part of the specification illustrate various forms and features of the present application; however, the present application should not be construed as being limited to those specific embodiments depicted in the drawings.
DETAILED DESCRIPTION
(19) For purposes of promoting an understanding of the principles of the invention, reference will now be made to the embodiments illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended, any alterations and further modifications in the illustrated device, and any further applications of the principles of the invention as illustrated therein being contemplated as would normally occur to one skilled in the art to which the invention relates.
(20) “Wastewater” typically includes one or more noxious and/or aromatic compounds and can be generated from a variety of sources. The wastewater processing systems and methods of the present application can be utilized to process wastewater which has been generated during the production of oil and/or gas. In addition to water itself, wastewater from oil and gas production is typically characterized by the presence of hydrocarbons (which can include crude oil as well as various lubricating oils, fats, and other oils), heavy metals, various volatile organic compounds, and other noxious and/or aromatic compounds.
(21) However, it is also contemplated that the teachings of the present application can be utilized to process agricultural wastewater, wastewater produced by various mining and mineral processing operations, landfill leachate wastewater, wastewater generated by healthcare facilities, wastewater generated by various manufacturing processes, agricultural wastewater, as well as other wastewater as would be known to a person of skill. “Unprocessed wastewater” or “raw wastewater” can refer to wastewater which has been generated but which has not yet been subject to processing and/or treatment.
(22) FIG. **1** is a schematic block diagram generally illustrating an exemplary wastewater processing system **100** in accordance with a first form of the present application. The wastewater processing system **100** will be described as processing raw wastewater **001** which has been generated by the oil and gas industry. It is believed that raw wastewater **001** produced by the oil and gas industry will typically not require filtration and/or screening prior to introducing the raw wastewater **001** into the wastewater processing system **100**. However, depending upon the type and size of solids present within the raw wastewater **001**, which may depend upon the source which generated the raw wastewater **001**, it may be desirable to utilize various filtration and/or screening processes (not shown) as are known, prior to introducing the raw wastewater **001** into the wastewater processing system **100**.
(23) The wastewater processing system **100** is depicted as including a water collection unit **200** which is in fluid communication with vaporization unit **300**. Raw wastewater **001** can be transferred into water collection unit **200** with a feed pump **101**. The raw wastewater **001** is transferred from the water collection unit **200** to be dispensed into vaporization unit **300**. The raw wastewater **001** can be continuously dispensed into the vaporization unit **300** with a pump **102**.
(24) A portion of the raw wastewater **001** is vaporized in the vaporization unit **300**, creating vapor **010**. The vaporization unit **300** utilizes heat to vaporize the portion of the raw wastewater **001**. In some forms, this vapor **010** can exit from the vaporization unit **300** to be released into the atmosphere exterior to the wastewater processing system **100**.
(25) As will be described herein, the vaporization unit **300** can take a number of forms, which include but are not limited to those forms described with regard to vaporization systems **009**, **011**, and/or **900**. The vaporization unit **300** can include an ignition source which is configured to interact with one or more of the hydrocarbons, volatile organic compounds, noxious compounds, and/or aromatic compounds present within the raw wastewater **001** and/or the vapor **010**. This ignition source can be flame **014**. The flame **014** is the combusting portion of an ignited fuel.

(26) It is believed that the introduction of flame **014** into the vaporization unit **300** can result in the ignition and combustion of some of the hydrocarbons, volatile organic compounds, and/or other noxious compounds within the vapor **010**. The incineration of the hydrocarbons and volatile organic compounds in the vapor **010** results in the production of exit vapor **012**. The exit vapor **012** can have a substantially reduced concentration of volatile organic compounds and hydrocarbons relative the vapor **010**. The exit vapor **012** is released into the atmosphere exterior to the wastewater processing system **100**.
(27) The flame **014** can also be the source of heat for the vaporization unit **300** (e.g., the flame **014** can generate the heat utilized to vaporize the portion of the raw wastewater **001**).
(28) A portion of the first aliquot of raw wastewater **001** that is not vaporized (i.e., un-vaporized wastewater **003**) collects in the vaporization unit **300**. In the vaporization unit **300**, the un-vaporized wastewater **003** continues to be exposed to heat. In various forms, hot air or the flame **014** may be directed toward the un-vaporized wastewater **003** and the hot air or flame **014** can contact a surface of the un-vaporized wastewater **003**, as will be described herein. The un-vaporized wastewater **003** can be heated and at least partially vaporized by the hot air or flame **014**. In one non-limiting form, the hot air or flame **014** provides sufficient heat to boil the un-vaporized wastewater **003** in the vaporization unit **300**. The absorption of heat by un-vaporized wastewater **003** can reduce the maximum temperature to which various components of the vaporization unit **300** are subjected to during operation.
(29) When the un-vaporized wastewater **003** reaches a threshold level within the vaporization unit **300**, the un-vaporized wastewater **003** can be transferred via a pump **103** into the water collection unit **200** where it at can at least partially mix with the raw wastewater **001** already present therein. The resulting combined wastewater **005** can be dispensed in a continuous stream from water collection unit **200** into the vaporization unit **300** via pump **102**, where some of the wastewater **005** is vaporized into vapor **010**. As is illustrated, raw wastewater **001**, un-vaporized wastewater **003**, and/or combined wastewater **005** can be dispensed into the vaporization unit **300**. As was discussed herein, vapor **010** can be released into the atmosphere or the vaporization unit can include an ignition source to convert vapor **010** into vapor **012** (e.g., by providing for the ignition and combustion of some of the hydrocarbons and volatile organic compounds therein).
(30) The vaporization unit **300** remains at substantially atmospheric pressure (e.g., substantially atmospheric pressure can range from around −15 PSIG to around 30 PSIG). The vaporization unit **300** preferably does not include a pressure vessel.
(31) The combined wastewater **005** that was not vaporized can collect in the vaporization unit **300** as un-vaporized wastewater **003**, where it can be further exposed to heat and recycled to the water collection unit **200**. The wastewater processing system **100** can continue to recirculate the wastewater **003**, **005** (e.g., transferring the un-vaporized wastewater **003** to the water collection unit and dispensing combined wastewater **005** into the vaporization unit **300**) until one or more of the following criteria are met: 1.) until the combined wastewater **005** or un-vaporized wastewater **003** contains a predetermined concentration of total dissolved solids TDS, total suspended solids TSS, hydrocarbons, volatile organic compounds VOCs, and/or other noxious compounds; 2.) until a predetermined amount of time has passed; 3.) until the source of available raw wastewater **001** is depleted; 4.) until only a minimum amount of combined wastewater **005** remains; or 5.) until other criteria are met as will occur to those skilled in the art. Preferably, the system **100** can be highly automated so that only minimal user intervention may be required during processing.
(32) The processed liquid waste **004** can then be removed from the system **100** for disposal. As will be appreciated to a person of skill, this processed liquid waste **004** has a substantially reduced overall volume than the volume of raw wastewater **001** which the processed liquid waste **004** was extracted from (e.g., as the vaporization unit **300** vaporized and removed much of the water from the raw wastewater **001**). This processed liquid waste **004** can include a higher concentration of TDS, TSS, heavy metals, and other noxious compounds, than the raw wastewater **001** (e.g., as the

concentration was diluted by the water present in the raw wastewater **001**).
(33) If additional wastewater is to be processed, another batch of raw wastewater **001** may be transferred into wastewater processing system **100** via water collection unit **200** for processing as described above. The further addition of raw wastewater **001** may be repeated as needed until the raw wastewater **001** to be processed is depleted.
(34) FIG. **2** depicts an exemplary water collection unit **200**. The water collection unit **200** can serve as a feed tank from which wastewater **001**, **005** can be supplied to the vaporization unit **300**. The water collection unit **200** collects raw wastewater **001** to be processed. The water collection unit **200** can receives aliquots of un-vaporized wastewater **003** from the vaporization unit **300** and can provide for the mixing of the un-vaporized wastewater **003** with raw wastewater **001** or combined wastewater **005** already present in water collection unit **200**. However, it is also contemplated that the un-vaporized wastewater **003** can be directly recirculated through the vaporization unit **300** (e.g., via pumping un-vaporized wastewater **003** from a fluid outlet **104** to a water distribution system **390** to be dispersed again within the vaporization unit **300**, See. FIGS. **3**A-**3**B).
(35) The exemplary water collection unit **200** includes a raw wastewater inlet **210** proximal to top portion **201**. The raw wastewater **001** may, for example, enter collection unit **200** through raw wastewater inlet **210** directly from a feed tank, from a gas or oil well, industrial processing site, landfill leachate pool, and/or naturally occurring water source.
(36) Aliquots of raw wastewater **001** to be processed are dispensed through the raw wastewater inlet **210** into the water collection unit **200** using any suitable method, including, but not limited to, via feed pump **101** as shown in FIG. **1**. Raw wastewater **001** may be dispensed through the raw water inlet **210** and into the water collection unit **200** at a rate suitable for the materials, equipment, and temperatures utilized. When the water collection unit **200** is utilized in connection with the vaporization system **011**, as will be described in connection with FIG. **3**B, raw wastewater **001** can be introduced into the water collection unit **200** a rate between about 6 and about 17 gallons per minute “gpm”.
(37) The water collection unit **200** can include a conical portion **202** located in the lower region **203**. This conical portion **202** aids in the collection of the heavier processed liquid waste **004**. In various embodiments, two water outlets **211**, **214**, a water inlet **212**, and a deflection plate **213** are located at the conical bottom portion **202**.
(38) Sensors **215** and **217** are depicted as being located at the water collection unit **200**. A controller **710** (see FIG. **7**) is placed in electronic communication with the sensors **215** and **217**. The sensor **215** and controller **710** can detect and/or determine the level of wastewater **001**, **005** present within the water collection unit **200** (e.g., the height of wastewater in the water collection unit **200**). The controller **710** can automatically manage the transfer of wastewater **001**, **003**, **005** (e.g., **001**, **003**, and/or **005**) passing into and out of the water collection unit **200**, at least in part as a function of the level detected.
(39) The sensor **217** and the controller **710** can detect and/or determine a current concentration of within the wastewater **001**, **005** present within the water collection unit **200**. For example, the sensor **217** and the controller **710** can detect and/or determine a current concentration of waste present within wastewater **001**, **005**. The waste can include total dissolved solids TDS, total suspended solids TSS, volatile organic compounds VOCs, and/or other noxious compounds which can be detected and/or determined with one or more sensors. The controller **710** can compare a current concentration of waste present within the wastewater **001**, **005** with a predetermined threshold waste concentration. In response to the current concentration being equal to or exceeding the threshold waste concentration, the controller **710** can determine to dispose of the wastewater **001**, **005** as processed liquid waste **004**. Additionally and/or alternatively, the sensor **217** and the controller **710** can detect and/or determine a concentration of water present within the wastewater **001**, **005**.
(40) The wastewater **001**, **005** is preferably continuously transferred from the collection unit **200** to

the vaporization unit **300** through a water outlet **211**. Similarly, the un-vaporized wastewater **003** is received into water collection unit **200** from the vaporization unit **300** through a water inlet **212** as described in further detail below.

(41) The water inlet **212** is preferably located below water outlet **211**. In locating the water outlet **211** above the water inlet **212**, less concentrated i.e., less dense, combined wastewater **005** (e.g., which may require additional processing) is readily dispensed from the water collection unit **200** to the vaporization unit **300** through the water outlet **211**, while more concentrated i.e., more dense, combined wastewater **005** will settle toward the bottom of the collection unit **200**.

(42) As has been described, the recirculation of un-vaporized wastewater **003** into water collection unit **200** from vaporization unit **300** and of wastewater **001**, **005** from the water collection unit **200** to the vaporization unit **300** can continue until one or more of the following criteria are met: 1.) until the combined wastewater **005** or un-vaporized wastewater **003** contains a predetermined concentration of one or more of total dissolved solids TDS, total suspended solids TSS, hydrocarbons, volatile organic compounds VOCs, and/or other noxious compounds; 2.) until a predetermined amount of time has passed; 3.) until the source of available raw wastewater **001** is depleted; 4.) until only a minimum amount of combined wastewater **005** remains; 5.) until the wastewater **001**, **005** in the water collection unit **200** exceeds a predetermine level (e.g., height in the tank); and/or 6.) until other criteria are met as will occur to those skilled in the art. The resulting processed liquid waste **004** is then released from the water collection unit **200** through the water outlet **214** using any suitable means e.g., such as via gravity feed, with a pump, or with another transfer device as will occur to a person of skill. The processed liquid waste **004** can be transferred through the water outlet **214** to be released to a storage tank, transport tank, or other destination, all as will occur to those skilled in the art.

(43) FIG. **3**A depicts a cross-sectional view of an exemplary vaporization system **009**. The vaporization unit **300** of wastewater processing system **100** can take the form of the vaporization system **009**. The vaporization system **009** includes a cylinder **301**. The cylinder **301** is at least partially defined by a top wall **303** and a bottom wall **304** which are connected by a contiguous wall **302**.

(44) The cylinder **301** is depicted as including an upper portion and a lower portion. The upper portion includes a vapor chamber **305** and the lower portion includes a separation chamber **306**. The vapor chamber **305** is located between the contiguous wall **302** of the cylinder **301** and walls **307** of the burner tube **360**. The vapor chamber **305** is located above the separation chamber **306** and can be separated from separation chamber **306** by a permeable substrate (e.g., which allows water and vapor to readily pass therethrough). An exemplary permeable substrate is grating **308**.

(45) As the vapor chamber **305** and the separation chamber **306** remain at substantially atmospheric pressure, the vapor chamber **305** and the separation chamber **306** do not need to be constructed or maintained as pressure vessels. Therefore, compliance with ASME and/or API pressure vessel codes should not be required for the vaporization system **009**.

(46) The phrase “at substantially atmospheric pressure” is intended to encompass a range of pressures from around −15 PSIG (negative fifteen pounds per square inch, gauge) to around 30 PSIG (thirty pounds per square inch, gauge). Such pressures are believed to be sufficiently close to atmospheric pressure as to not require a pressure vessel, as would be understood to a person of skill. In one non-limiting form, the vapor chamber **305** and the separation chamber **306** can operate at pressures in the range of 0 PSIG to 7 PSIG.

(47) A burner **350** can serve as a heat source for the vaporization system **009**. The burner **350** is disposed proximal to the top wall **303** of the cylinder **301**. A burner tube **360** extends from the burner **350** through the top wall **303** of cylinder **301**, into and through the vapor chamber **305**, through the grating **308** and partially into the separation chamber **306**. The burner **350** receives and ignites a fuel. The ignited and combusting fuel, and the hot exhaust gas, are expelled into the burner tube **360**.

(48) A variety of gaseous and liquid fuels can be utilized by burner **350**. It is believed that gaseous fuels such as propane, methane, and natural gas (including flare gas), may be preferable for many applications due to a ready availability thereof. The burner **350** can take a variety of forms as will occur to a person of skill in the art, including as a dual or multiple fuel burner **350** (e.g., which can combust different gaseous and/or liquid fuels) or as a single or dedicated fuel burner **350**.
(49) A blower **370** is configured to provide air to the burner **350** (e.g., enabling combustion) and to blow the exhaust gases and hot air (depicted in FIG. **3**A as downward arrows "D") from the burner **350** into and through the burner tube **360**. The exhaust gases and hot air are expelled from the burner tube **360** out into separation chamber **306**. In some embodiments, the blower **370** is from PENNBARRY of Plano, Tex. USA and has a "silencer" built in to reduce the amount of sound produced. However, a variety of blowers **370** may be utilized herein, as would be understood to a person of skill. The force of blower **370** causes heated exhaust gases to deflect from a bottom wall **304** of the separation chamber **306** or from any un-vaporized wastewater **003** that has collected there, to rise up through grating **308** and through the vapor chamber **305** (depicted in FIG. **3**A as upward arrows, labeled "U").
(50) Raw wastewater **001**, un-vaporized wastewater **003**, or combined wastewater **005** can be introduced into the vaporization unit **300** through one or a plurality of water inlet ports **380** proximal to the top of the vaporization unit **300**. This wastewater **001**, **003**, **005**, passes through the water distribution system **390** that disperses the wastewater inside vapor chamber **305**. The wastewater **001**, **003**, **005** which is dispersed into the vapor chamber **305** is depicted as falling wastewater droplets **20**. The water distribution system **390** will be described herein in connection with FIGS. **4**A-**4**C.
(51) The burner tube **360** extends from burner **350** through top wall **303** of cylinder **301**, into and through vapor chamber **305**, through grating **308**. As the burner tube **360** is configured to transport heated exhaust and combustion air gases (depicted in FIG. **3**A as downward arrows "D"), the burner tube **360** is preferably constructed from a suitable high temperature material, and it may be desirable to provide various safeguards to prevent the burner tube **360** from deteriorating and/or failing due to the repeated exposure to alternating temperatures, thus experienced repeated occurrences of the materials forming burner tube **360** expanding and contracting, as well as vibrations associated with operation of blower **370** and other components of the system **100**.
(52) The wastewater processing system **100** vaporizes wastewater **001**, **003**, **005**, by exposing the wastewater **001**, **003**, **005** to heat within the wastewater processing system **100**. For example, with regard to the vaporization system **009** of FIG. **3**A, heated exhaust gases traveling through burner tube **360** heat the walls **307** of burner tube **360**. The wastewater droplets **020** which contact or pass close to the walls **307** of the burner tube **360** may absorb heat therefrom.
(53) The wastewater droplets **020** falling through the vapor chamber **305** and falling down through the separation chamber **306** may absorb heat from heated exhaust gas (depicted in FIG. **3**A as upward arrows, labeled "U") which has deflected from the bottom wall **304** of the separation chamber **306**, and/or has deflected from any un-vaporized wastewater **003** that has collected in the separation chamber **306**, and that rises up through the grate **308** and through the vapor chamber **305**.
(54) The un-vaporized wastewater **003** collected in the separation chamber **306** may absorb heat from heated exhaust gas that is exiting the burner tube **360**. This heated exhaust gas can contact the surface of un-vaporized wastewater **003** (downward-moving exhaust gas being depicted in FIG. **3**A as downward arrows, labeled "D"). Exposure of wastewater **001**, **003**, **005**, to heat at one or more of these locations causes at least a portion of the wastewater **001**, **003**, **005** to vaporize into vapor **010**. Vapor **010** can exit the vaporization unit **300** to pass into the atmosphere through one or more vents **340** (e.g., depicted as two vents **340** in FIG. **3**A) and stacks **361**, **362**. In some embodiments, an exhaust fan is included within or adjacent to the at least one vent **340** to assist with drawing vapor **010** out of the vent **340**.

(55) As will be understood to a person of skill, the term "chamber" refers to a cavity or space which is at least partially enclosed. For example, as has been described herein the separation chamber **306** and the vapor chamber **305** are not fully enclosed; rather, the separation chamber **306** is in flow communication with the vapor chamber **305**, the burner tube **306** is in flow communication with the separation chamber **306**, and the vapor chamber **305** is in flow communication with the atmosphere (e.g., via the vent **340**).

(56) FIG. 3B depicts a cross-sectional view of another exemplary vaporization system **011**. The vaporization unit **300** of wastewater processing system **100** can take the form of the vaporization system **011**. The vaporization system **011** is depicted as an alternative form to the vaporization system **009**; however, it is contemplated that various components, features, and/or subunits from vaporization systems **300**, **009**, **011**, and **900** can be interchanged, combined, removed, and/or modified in a variety of manners, as would be understood to a person of skill. The vaporization system **011** can include a similar structure to the vaporization system **009** and can include similar features as the vaporization system **009**, except as will be set forth herein.

(57) The vaporization system **011** includes an ignition source which is configured to provide for the ignition of volatile compounds and/or hydrocarbons within the chamber **042**. The volatile compounds can be volatile organic compounds. In the vaporization system **011**, the ignition source is depicted as taking the form of flame **014**, which is introduced into the chamber **042**. The flame **014** can interact with the vapor **010** (e.g., the vaporized wastewater), the falling wastewater droplets **020**, and/or the collected un-vaporized wastewater **003** within the chamber **042**.

(58) It has been discovered, through highly preliminary testing of the vaporization system **011**, that the introduction of the flame **014** into the chamber **042** can result in a reduction of volatile organic compounds and hydrocarbons which are released to the atmosphere in exit vapor **012**. It is believed that this reduction results from the ignition, combustion, and incineration of at least some of the hydrocarbons and/or volatile compounds which are present within the vapor **010**. The destruction of some of the hydrocarbons and/or volatile compounds present within the vapor **010** (e.g., via ignition, combustion, and incineration) provides exit vapor **012**.

(59) The vaporization system **011** is depicted as taking the form of a cylinder **301**. The cylinder **301** is defined by a contiguous wall **302** which connects a top wall **303** of cylinder **301** to bottom wall **304** of cylinder **301**. A chamber **042** is at least partially defined by the contiguous wall **302** of the cylinder.

(60) The chamber **042** includes an upper region and a lower region. The upper region can take the form of a vapor chamber **021** and the lower region can take the form of a collection chamber **040**. The vapor chamber **021** is depicted as being located between the contiguous wall **302** of the cylinder **301** and the walls **007** of a burner tube **013**. The collection chamber **040** is depicted as being located below the vapor chamber **021**.

(61) The vapor chamber **021** can be separated from collection chamber **040** by a permeable substrate which allows water and vapor to readily pass therethrough. This permeable substrate is depicted as a grating **308**; however, a variety of permeable substrates, including various meshes, are contemplated herein.

(62) Similar to vaporization system **009**, the vaporization system **011** utilizes heat to vaporize a portion of the dispensed wastewater **001**, **003**, **005**. The wastewater **001**, **003**, **005**, can be dispensed into the system via the water distribution system **390** or through other fluid introduction systems as would be known to a person of skill.

(63) The flow of wastewater **001**, **003**, **005** through the exemplary vaporization system **011** will now be described. The water distribution system **390** can be utilized to expel wastewater droplets **020** into the vapor chamber **021**. The force of gravity moves the wastewater droplets **020** downwardly through the vapor chamber **021** and the collection chamber **040**. As will be described hereinafter, the wastewater droplets **020** can contact and interact with a vaporization medium **017** (e.g., pall rings **500**) located in the vapor chamber **021**.

(64) Wastewater droplets **020** which were not vaporized in the vapor chamber **021** will pass through the grating **308** and will fall downwardly into the collection chamber **040**. Some of the wastewater droplets **020** will vaporize (e.g., into vapor **010**) as they move downwardly through the collection chamber **040**. Wastewater droplets **020** which remain un-vaporized as they travel through the collection chamber **040** collect above the bottom wall **304** of the collection chamber **040** as un-vaporized wastewater **003**.
(65) The vaporization system **011** includes a burner, depicted as burner **350**, and a blower, depicted as blower **370** (as were described with regard to FIG. **3**A). It is contemplated that the burner **350** and blower **370** can take a variety of forms through which flame **014** can be discharged into the collection chamber **040**.
(66) The burner tube **013** extends from the burner **350** through the top wall **303** of cylinder **301** and can extend through the vapor chamber **021**. Flame **014** is expelled from a discharge **016** at a distal end **022** of the burner tube **013** into the collection chamber **040**. The distal end **022** of the burner tube **013** can extend through the grating **308** and into the collection chamber **040**.
(67) The burner **350** provides an ignited fuel into the burner tube **013**. The blower **370** blows the ignited fuel through the burner tube **013**, out of the discharge **016**, and into the collection chamber **040**. The blower **370** also serves to provide air for the combustion of the fuel. The vaporization system **009** and the vaporization system **011** are both depicted as utilizing burner **350** and blower **370**; however, various control parameters for the burner **350** and the blower **370** may need to be adjusted and/or altered to discharge the flame **014** into the collection chamber **040** (e.g., the fuel flow rate to the burner **350**, the air flow rate of the blower **370**, and other control parameters as would be understood to a person of skill).
(68) Preferably, the flame **014** can serve as the source of heat for the vaporization system **011** (e.g., to vaporize a portion of the dispensed wastewater **001**, **003**, **005**) as well as serving as the ignition source (e.g., to ignite and combust one or more hydrocarbons and/or volatile organic compounds). However, it is also contemplated that a separate ignition source and heat source may be utilized (e.g., exemplary heat sources can include various burners, resistance heaters, heat pumps, etc. and exemplary ignition sources can include various spark, flame **014**, or other ignition sources as are known).
(69) Referring now to FIGS. **3**B and **8**, an exemplary flame **014** flow path through the vaporization system **011** will now be described. FIG. **8** is an enlarged view of the area **031** from FIG. **3**B, which depicts the flame **014** in the collection chamber **040** as during operation. It is to be understood that although specific temperatures and ranges are described hereinafter, such temperatures and ranges are solely intended to be exemplary and are to be construed as non-limiting. The use of a variety of temperatures and/or temperature ranges is contemplated herein. Additionally, although an exemplary flow path of the flame **014** will be discussed hereinafter, it is contemplated that the ignition source (e.g., such as flame **014**) can be disposed in a variety of locations in the vaporization system **011** to provide for the combustion of at least some of the hydrocarbons and/or volatile compounds therein.
(70) The flame **014** is expelled from the discharge **016** into the collection chamber **040**. The flame **014** can include a temperature in the range of 500° F. to 3500° F. when exiting the discharge **016**. The flame **014** can include a temperature in the range of 1000° F. to 2500° F. when exiting the discharge **016**. In one specific form, the flame **014** can include a temperature in the range of 1600° F. to 1800° F. at the discharge **016**.
(71) Preferably, the collection chamber **040** serves as an expansion chamber **018** for the flame **014**. The collection chamber **040** can include a sufficiently large internal volume for the ignited and combusting fuel (e.g., generating the flame **014**) to expand and fully combust (e.g., interacting with the air supplied by the blower **370**) within the collection chamber **040**, resulting in the conversion of the fuel and flame **014** into heated exhaust gas. The blower **370** quickly propels the flame **014** thorough the burner tube **013** to the collection chamber **040** for the fuel to be fully combusted. In

one non-limiting form, the blower **370** can pressurize the burner tube **013**.
(72) Preferably, the flame **014** contacts an upper surface **015** of the un-vaporized wastewater **003**. This contact with the upper surface **015** can serve to direct the flame **014** outwardly toward an inner wall **019** of the collection chamber **040** (as best shown in FIG. **8**). The flame **014** contacts and interacts with the upper surface **015** of the un-vaporized wastewater **003**, and heat is absorbed from the flame **014** by the un-vaporized wastewater **003**.
(73) The flame **014** flows upwardly toward the vapor chamber **021**. Some of the flame **014** can contact the inner wall **019** of the collection chamber **040**, to then be directed by the inner wall **019** toward the vapor chamber **021**. If the flame **014** has a temperature in the range of about 1600° F. to about 1800° F. at the discharge **016**, the flame **014** can include a temperature in the range of about 900° F. to about 1100° F. after contacting the upper surface **015** when rising (e.g., depicted at **029**) toward the vapor chamber **021**. The drop in flame **014** temperature is believed to be due in part to the amount of heat absorbed by the un-vaporized wastewater **003**.
(74) Although it is contemplated that the flame **014** can enter the vapor chamber **021**, preferably the combusting fuel (e.g., generating the flame **014**) is fully combusted in the collection chamber **040**. In this manner, heated exhaust gases produced by flame **014** will rise upwardly into the vapor chamber **021** flowing through the vaporization medium **017**.
(75) The vapor chamber **021** and the collection chamber **040** remain at substantially atmospheric pressure during operation of the vaporization system **011**. This is believed to be highly advantageous as maintaining the vapor chamber **021** and the collection chamber **040** at substantially atmospheric pressure can eliminate the need for the chamber **021**, **040** to be constructed as pressure vessels.
(76) As will be appreciated to a person of skill, the vapor chamber **021** and the collection chamber **040** can be maintained at substantially atmospheric pressure by providing sufficient flow out of the system to enable a mass flow rate out of the vaporization system **011** to equal the mass flow rate into the vaporization system **011**. For example, the vaporization system **011** can include one or more of the following features: the burner tube **013** can have a smaller flow area than a flow area of the vapor chamber **021**; the collection chamber **040** can include a sufficiently large volume to serve as the expansion chamber **018** for the flame **014**; the vent(s) **340** in flow communication between the vapor chamber **021** and the atmosphere can include a large flow area relative to the flow area of the burner tube **013**; and the volume of air supplied by the blower **370** and the amount of fuel provided by the burner **350** can be considered and accounted for when designing the various flow areas of the vaporization system **011**.
(77) During operation, wastewater **001**, **003**, **005**, is introduced into the vapor chamber **021** as wastewater droplets **020**. The droplets **020** can be sprayed at any vaporization medium **017** in the vapor chamber **021**. The droplets will pass downwardly through the vapor chamber **021**, and through any vaporization medium **017**, via gravity.
(78) The vapor chamber **021** receives heat from the flame **014** which transfers through the walls **007** of the burner tube **013** and heat from the exhaust gases and vapor **010** as they rise upwardly through the vapor chamber **021**. The heated exhaust gases and vapor **010** flow around and/or through any vaporization medium **017** (e.g., pall rings **500**) located in the vapor chamber **021**, heating the pall rings **500**.
(79) The wastewater droplets **020** can interact with the pall rings **500** (e.g., some wastewater droplets **020** being vaporized upon contact with the pall rings **500**, other wastewater droplets **020** contacting and flowing on an outer surface of the pall rings), and the pall rings **500** can serve to increase the amount of time the falling wastewater droplets **020** spend within the vapor chamber **021** and can introduce turbulence into the upward flow of heated exhaust gases and vapor **010**.
(80) The pall rings **500** may be at a lower temperature than the exhaust gas and vapor **010**, which flow upwardly therethrough. Heat is transferred to the pall rings **500** from the exhaust gas, vapor **010**, and can be transferred from the walls **007**. Heat is transferred from the pall rings **500** to the

wastewater droplets **020**. A temperature at an upper and outer portion **027** of the pall rings **500** can be lower than a temperature at an inner and lower portion **028** of the pall rings **500** (e.g., the pall rings **500** which are further away from the expansion chamber **018** and the walls **007** can have a lower temperature). In many applications, the temperature of the incoming wastewater **001**, **003**, **005** will be less than the exhaust gas temperature in the vapor chamber **021**; therefore, the water droplets **020** will cool the pall rings **500** as heat transfers to the water droplets **020**.
(81) With regard to the prior example where flame **014** is introduced into the collection chamber **040** at a temperature in the range of about 1600° F. to about 1800° F., a centrally located surface **026** of the pall rings **500** can have a temperature in the range of about 250° F. to about 400° F. It is important to note that this is not necessarily the temperature of the exhaust gas and/or vapor **010** at this location; rather, this is a surface **026** temperature of a centrally located pall ring **500**. The use of pall rings **500** is believed to provide an increased rate of heat transfer to the wastewater droplets **020**, relative a vapor chamber **021** without vapor medium **017** therein.
(82) It is believed that a substantial portion of the wastewater droplets **020** are vaporized, and turned into vapor **010**, in the vapor chamber **021**. It is believed that 5% or more of the wastewater **001**, **003**, **005** introduced to the vaporization system **011** may vaporize in the vapor chamber **021**. The crossflow action provided within the vapor chamber **021** is believed to increase the vaporization of the wastewater droplets **020** (e.g., heated exhaust gas flows upwardly in an opposite direction relative to the downwardly moving water droplets **020**).
(83) Wastewater droplets **020** that were not vaporized in the vapor chamber **021** will pass through the grating **308** into the collection chamber **040**. As is illustrated in FIG. **8**, a portion of the un-vaporized droplets **020** can pass through the flame **014** rising up from the surface **015** of the un-vaporized wastewater **003**. The contact of these un-vaporized droplets **020** with the flame **014** can result in the vaporization of some of the droplets **020** and the production of vapor **010**. Additionally, the flame **014** may combust some of the solids or other noxious compounds within the droplets **020**. Any droplets **020** which remain un-vaporized as they pass downwardly through the collection chamber **040** will collect on the bottom wall **304** as un-vaporized wastewater **003**.
(84) As was previously discussed, the flame **014** preferably contacts the surface **015** of the un-vaporized wastewater **003**. The un-vaporized wastewater **003** can absorb a substantial amount of heat through this flame **014** contact and may boil inside the collection chamber **040**. The combination of flame **014** contact and boiling of the un-vaporized wastewater **003** vaporizes a portion of the un-vaporized wastewater **003**, and it is likely that various solids and/or noxious compounds within the un-vaporized wastewater **003** may be incinerated by this flame **014** contact, especially those which rise toward the surface **015**.
(85) The vaporization system **011** vaporizes a substantial amount of the wastewater **001**, **003**, **005** introduced into the vaporization system **011**. The vapor **010** resulting from the vaporization of wastewater **001**, **003**, **005** will include a large concentration of water vapor; however, this vapor **010** can also include various undesirable vapors, including volatile organic compounds, various vaporized hydrocarbons, aromatics, and/or other noxious compounds. As has been described herein, the flame **014** introduced into the collection chamber **040** serves as an ignition source for undesirable combustible vapors (e.g., volatile organic compounds, vaporized hydrocarbons, and other combustible noxious compounds) and the undesirable combustible vapors can be incinerated within the vaporization system **011**.
(86) Two vents, depicted as stack **361** and stack **362**, are in flow communication with the vapor chamber **021**. The exit vapor **012**, which has been discovered to be low in volatile organic compounds, can pass through the stacks **361**, **362**, through the outlets **030** and into the atmosphere.
(87) FIGS. **4**A-**4**C depict an exemplary water distribution system **390** which can be utilized within the vaporization systems **009**, **011** as have been described herein. FIG. **4**A depicts a front view of a water distribution subunit **400** of the water distribution system **390**. The water distribution subunit **400** includes a pipe **401** that is fluidly connected with a water inlet port **380**. The pipe **401** is

perforated with a plurality of holes **402**. The pipe **401** has open ends **403**. In use, wastewater **001**, **003**, **005** enters the pipe **401** through the water inlet port **380**, flows through the pipe **401**, and exits the pipe **401** through the plurality of holes **402** and/or through the open ends **403** of pipe **401**. The wastewater droplets **020** then fall downwardly into and through the vapor chamber **305** (e.g., due to the force of gravity acting upon the wastewater droplets **020**).

(88) FIG. **4**B is a bottom plan view of the water distribution subunit **400**. The subunit **400** may comprise a pipe **401** that is bent to mimic or approximate the inside curvature of the contiguous wall **302** of the cylinder **301**. Alternative configurations of the water distribution system **390** to introduce wastewater **001**, **003**, **005** into the vapor chamber **305**, preferably as wastewater droplets **020**, will occur to those skilled in the art in view of this disclosure.

(89) FIG. **4**C depicts a top cross-sectional view looking down into the vaporization unit **300**, which is depicted as taking the form of vaporization system **009**. A series of water distribution subunits **400**, such as the one shown in FIG. **4**B, may be distributed around the inside of the contiguous wall **302** of the cylinder **303** in order to distribute wastewater **001**, **003**, **005** fairly evenly inside the vapor chamber **305**.

(90) Referring now to FIGS. **3**A-**3**B and **5**, to prolong the exposure time of the wastewater droplets **020** to the heated exhaust gases in the vapor chambers **305**, **021** a vaporization medium **017** can be placed within the vapor chambers **305**, **021**. This vaporization medium **017** can also serve to provide a large, heated surface area, heated by the exhaust gases rising upwardly, for the wastewater droplets **020** to contact and absorb heat from.

(91) This vaporization medium **017** can take the form of one or more baffles. As is illustrated in FIGS. **3**A-**3**B, the vaporization medium **017** can take the form of a plurality of pall rings **500**. The plurality of pall rings **500** can be located in the vapor chambers **305**, **021** and can at least partially fill the vapor chambers **305**, **021**. One exemplary pall ring **500** is depicted in FIG. **5**. This pall ring **500** is a CHRISTYPAK metal pall ring distributed by Christy Catalytics, of St. Louis, Mo. This exemplary pall ring **500** is constructed of 304 stainless steel and has a nominal outer diameter of 1.5″. This exemplary pall ring **500** is a substantially hollow cylinder which has a plurality of openings **501** and protrusions **502**.

(92) When located in the vapor chambers **305**, **021**, these pall rings **500** can absorb and retain heat from the heated exhaust gas passing through the vapor chambers **305**, **021**, can induce turbulence in the heated exhaust gas, and can greatly increase the heated surface area to which the wastewater droplets **020** passing therethrough are exposed. This heat retention, turbulence, and increased heated surface area are believed to promote more efficient heat transfer between the heated exhaust gas and the wastewater droplets **020**, which may result in an increased vaporization of the wastewater droplets **020**.

(93) However, even with the increased turbulence and heat transfer provided by the pall rings **500** in the vapor chambers **305**, **021**, only a portion of wastewater **001**, **003**, **005** (which is introduced as wastewater droplets **020**) is vaporized into vapor **010** as it passes downwardly through the vapor chambers **305**, **021**. For example, in some embodiments in which about 50 gpm of combined wastewater **005** is introduced into the vaporization unit **300**, only about 6 gpm to 15 gpm of wastewater may be converted into vapor **010**. However, as would be appreciated to a person of skill, the vaporization of about 5% to 10% or more of wastewater **001**, **003**, **005** per cycle through the vaporization chambers **305**, **021** is a substantial portion of the wastewater **001**, **003**, or **005**.

(94) The un-vaporized wastewater **003** can collect in the collection chamber **040** of the vaporization system **011** (FIG. **3**B) or in the separation chamber **306** of the vaporization system **009** (FIG. **3**A). The presence of un-vaporized wastewater **003** in the collection chamber **040** or separation chamber **306** may provide a number of benefits which include, but are not limited to, serving as a heat shield for the bottom wall **304**, enabling the collection and concentration of TDS, TSS, etc. for subsequent processing or disposal, and providing an additional opportunity for the un-vaporized wastewater **003** to be vaporized and/or incinerated (e.g., via the flame **014** directly

contacting the un-vaporized wastewater **003** as shown in FIG. **3**B).
(95) Advantageously, the wastewater processing system **100** in accordance with the present disclosure vaporizes wastewater **001**, **003**, **005** without the need to employ pressure vessels and/or to pressurize the vaporization unit **300**. Consequently, the vaporization unit **300** may operate at substantially atmospheric pressure, thereby eliminating the dangers, economic cost, and inconvenience that might be associated with pressure vessels.
(96) With continued reference to FIGS. **3**A-**3**B, the level of un-vaporized wastewater **003** in the separation chamber **306** (or the collection chamber **040**) can be monitored using an automated sensor **330**. Once the level of un-vaporized wastewater **003** is above a predetermined depth, a controller **710** (see FIG. **7**) can automatically activate a pump (e.g., **103**) to transfer excess un-vaporized wastewater **003** from separation chamber **306** (or the collection chamber **040**) back to the water collection unit **200** via water inlet **212** for subsequent processing or disposal.
(97) As the un-vaporized wastewater **003** is heated while it is in vaporization unit **300** as has been described above, when the un-vaporized wastewater **003** is returned to the water collection unit **200** and mixed with raw wastewater **001** or combined wastewater **005** already present in water collection unit **200**, further heat transfer occurs. The mixing of the un-vaporized **003** wastewater into the collection unit **200** will thus raise the temperature of the next aliquot of combined wastewater **005** to be transferred to and discharged into the vaporization unit **300**, which can increase the vaporization efficiency in further cycles.
(98) As has been discussed, when sufficient raw wastewater **001** or combined wastewater **005** has been vaporized, when or sufficient time has passed, or for other reasons understood by those skilled in the art, the wastewater processing system **100** may be shut down, either automatically via the controller **710** or through manual user control. Depending on the implementation and/or context, wastewater **001**, **003**, **005** may cease to be delivered to the vaporization unit **300**, and the vaporization unit **300** may continue to operate the burner **350** and the blower **370** for a period of time while the wastewater droplets **020** pass through the vapor chamber **305**, **021** down to the separation chamber **306** (FIG. **3**A) or the collection chamber **040** (FIG. **3**B). The burner **350** and the blower **370** can then be turned off, and the un-vaporized wastewater **003** which has collected in separation chamber **306** (FIG. **3**A) or the collection chamber **040** (FIG. **3**B) can be transferred from the fluid outlet **104** back to the water collection unit **200** via the water inlet **212**.
(99) The vaporization system **009**, shown in FIG. **3**A, comprises a cleaning port **315** in the separation chamber **306**. If, for example, un-vaporized wastewater **003** present in the separation chamber **306** becomes saturated with other materials, e.g., if chloride and solids precipitate from the un-vaporized wastewater **003**, unwanted deposits may occur inside the separation chamber **306**. In these and other circumstances, the cleaning port **315** may be useful to allow ready access into the separation chamber **306** for cleaning and other purposes. The collection chamber **040** of the vaporization system **011** can include a cleaning port (not shown) if desired.
(100) FIG. **6** is a top plan view of an exemplary water transfer system **600** for transferring raw wastewater **001**, un-vaporized wastewater **003**, and combined wastewater **005** between the water collection unit **200** and the vaporization unit **300**. The exemplary water transfer system **600** can be used in connection with the mobile processing system **1000**, as will be described in connection with FIG. **7** below.
(101) The exemplary water transfer system **600** comprises a series of hoses and/or pipes **601**, **602**, **603**, **604** and pumps **101**, **102**, **103**, and **613**. Raw wastewater **001** is pumped from a storage location by a pump **101**, through pumping **604**, and through raw wastewater inlet **210** into the collection unit **200**. Raw wastewater **001**, un-vaporized wastewater **003**, or combined wastewater **005** is dispensed from the collection unit **200** through the water outlet **211**, through the plumbing **601** where it is acted upon by the pump **102**, and through the water inlet **380** into the vaporization unit **300**.
(102) The un-vaporized wastewater **003** is transferred from the vaporization unit **300** through the

fluid outlet **104**, through the plumbing **602** where it is acted upon by the pump **103**, and through the water inlet **212** into the collection unit **200**. Processed liquid waste **004** is removed from the collection unit **200** for storage and/or disposal through the water outlet **214**, through plumbing **603** via pump **613**. Although a specific water transfer system **600** has been described, it is contemplated that wastewater **001**, **003**, **005** can be transferred to/from the vaporization unit **300** through a variety of pumps, gravity feeds, or other transfer devices as are known.
(103) FIG. **7** is a schematic cross-sectional view of an exemplary mobile wastewater processing system **1000**. This mobile wastewater processing system **1000** is depicted as being located on a trailer **1200**. The mobile wastewater processing system **1000** includes the vaporization unit **300**, burner **350** and blower **370**, controller **710** (which can have a controls panel incorporated thereto), can include collection unit **200**, and may include water transfer system **600**.
(104) The mobile wastewater processing system **1000** can be enclosed within a housing **703**. The housing **703** can include a top panel **701**, a front panel **700**, a rear panel **702**, and two side panels (not shown). Stacks **361**, **362** of the vents **340** can exit through the top **701** of the housing **703**. Vapor **010** or **012** exists the vaporization unit **300** through the stacks **361**, **362** and passes into the atmosphere. In some implementations, stacks **361**, **362** are “no-loss” stacks that collect heavy compounds, condensate water, and other liquids they encounter into collection unit **200** (as were described in FIG. 8 of U.S. application Ser. No. 17/093,511). However, in further forms, stacks **361**, **362** simply allow such materials to escape with the vapor **010** or **012**.
(105) In the example shown in FIG. **7**, the blower **370** is fluidly connected to a ventilation pipe **391**, which extends through the rear panel **702** of the housing **703**, where it is exposed to the atmosphere outside of the housing **703**. The ventilation pipe **391** is configured to feed the blower **370** with air from the atmosphere.
(106) The housing **703** is depicted as being is disposed on the bed of a commercially available trailer **1200** that may be hauled by a pickup truck; however, it is contemplated that the mobile wastewater processing system **1000**, can be mounted on a variety of towable trailers, skids, or other mobile platforms as are known. Advantageously, the mobile wastewater processing system **1000** may be delivered directly to a remote location, such as the site of an oil or gas well or other source of wastewater, where it may be used to process the wastewater as has been described.
(107) Vapor **010** or **012**, depending upon if an ignition source such as flame **014** is included in the vaporization unit **300**, may be released from wastewater processing system **1000** on site, thus eliminating the requirement to store and/or haul much of the produced wastewater away from the site. By vaporizing the wastewater and releasing the water vapor, raw wastewater **001** may be reduced in volume by up to approximately 95%, leaving only the processed liquid waste **004** in volumes as low as 5% (e.g., 1/20.sup.th) of the original volume of wastewater **001** to be disposed of by conventional measures. Once the wastewater processing at a given site is completed, the wastewater processing system **1000** may subsequently be moved to other sites to process wastewater as needed.
(108) The housing **703** can include a top panel **701**, a front panel **700**, a rear panel **702**, and two side panels (not shown). These panels **701**, **700**, **702**, and the two side panels can be constructed of stainless steel sheets on the interior with a one-inch polystyrene insulation layer and exterior panels of 18-gauge Kynar-coated galvanized steel. The housing **703** can be approximately 8 feet wide, 24 feet long, and 13 feet high 8′×24′×13′; however, the housing **703** can be more compact, especially if the collection unit **200** is not desired. The small overall dimensions of the mobile wastewater processing system **1000** facilitates easier transport and use in numerous applications, especially those in remote locations. However, it is contemplated that the mobile wastewater processing system **1000** can utilize a variety of materials, coatings, and dimensions, and can take a variety of configurations, as will occur to those skilled in the art.
(109) Electrical and electromechanical components of the wastewater processing systems **100**, **1000** may be connected to and controlled by a control system. The control system comprises

modules to functionally execute steps of processing of raw wastewater **001**. The control system may include an operating conditions module, a user request module, a processing control module, an actuation module, and other modules known to those skilled in the art.

(110) The operating conditions module interprets operating conditions, such as temperatures, pressures, water and gas flow rates, presence of unwanted gases, e.g., carbon dioxide, hydrogen sulfide, and the like, by reading various sensor outputs, receiving data messages over a network, and/or through any other method understood in the art. The user request module interprets at least one user request, which, in the present embodiment, includes a target parameter value, for example, a processing target in terms of weight, volume, or other metric or time target. The user request module may process the user request by sending a data message over a network, by comparing an operating condition to a pre-calibrated value, or the like. Without limitation, a user of the request module may be a person, but may also be an application, a computer in communication with the application, and/or a computer in use by a person not shown monitoring the system, and the request may be produced in a direct response to an action by the user, may be scheduled, or may otherwise be programmatically generated as will occur to those skilled in the art.

(111) The control system may further include a network that connects more than one computer to each other and to the control system. Additionally, the network may connect modules within the control system to each other and may connect the control system to the wastewater processing systems **100**, **1000**. The network may comprise multiple components, for example a LAN, WAN, satellite connection, the internet, and communications within a server and/or computer. Since the control system may allow for the wastewater processing systems **100**, **1000** to be controlled remotely, it may eliminate the need for having personnel physically present to operate wastewater processing operations.

(112) The wastewater processing systems **100**, **1000** according to the present disclosure may be placed upon a variety of mobile platforms, such as shown in FIG. **7**. For example, mobile wastewater processing systems **100**, **1000** may placed on a skid or suitable frame structure to be airlifted to remote locations via helicopter or cargo plane. Thus, the wastewater processing systems **100**, **1000** may be delivered directly to where raw wastewater **001** is generated, including but not limited to the site of an oil or gas well, industrial processing facility, landfill leachate pool, livestock wastewater holding ponds, and/or other wastewater generating source. Consequently, wastewater processing systems **100**, **1000** may eliminate or substantially reduce the need to ship wastewater from sites, and can be especially advantageous at remote sites, thus eliminating or simplifying often complex logistical issues and the associated costs.

(113) In various embodiments, the wastewater processing systems **100**, **1000** are powered by fuel present at the production site. For example, electrical power can be provided to the wastewater processing systems **100**, **1000** via a generator that operates on either the gas that is being generated at the site or a byproduct of this gas generation (e.g., either natural gas, propane, methane, and/or flare gas), and the fuel used by the burner **350** can be the gas that is being generated at the site or a byproduct of this gas generation.

(114) In various embodiments, the burner **350** is adapted to use any of a plurality of different types of fuel. In one exemplary form, the burner **350** can take the form of a 10 MM BTU MAXON OVENPAK LE EB100 Natural Gas Burner with standard mixing cone, short flame type 310 stainless steel burner sleeve, direct spark with flame rod ignition, all from Honeywell Muncie, Ind. However, a variety of burners **350**, including multi-fuel burners and single-fuel burners, may be used in other embodiments as will occur to those skilled in the art.

(115) In various embodiments, the sensors **215** and **330** are radar sensors such as the VEGAPULS 64 distributed by VEGA Grieshaber KG Schiltach, Germany. In other embodiments, other radar sensors or various other types of proximity sensors can be utilized to monitor the levels of wastewater **001**, **005** in the bottom of the water processing unit **200** and the level of un-vaporized wastewater **003** in the vaporization unit **300**.

(116) The sensor **217** and the controller **710** can detect and/or determine a current concentration of waste present within wastewater **001**, **005** within the water processing unit **300**. Alternatively, the sensor **217** can be located in the vaporization unit **300** so that it can interact with the un-vaporized wastewater **003** therein. It is contemplated that the sensor **217** can take the form of a conductivity sensor (e.g., to detect TDS or TSS within the wastewaters **001**, **003**, **005**), can take the form of a clarity sensor, and/or can be a specified sensor to detect various volatile organic compounds and/or hydrocarbons. The sensor **217** can also comprise a sensor array.
(117) FIG. **9** illustrates yet another alternative embodiment of the vaporization unit **300**, which is depicted as vaporization system **900**. The vaporization system **900** includes the same features and functions as the vaporization systems **009**, **011**, unless otherwise described, and may be utilized as the vaporization unit **300** in the wastewater processing systems **100**, **1000**.
(118) The vaporization system **900** includes a cylinder **901** defined by a contiguous wall **902** which connects a top wall **903** of the cylinder **901** to a bottom wall **904** of the cylinder **901**. The cylinder **901** comprises two internal chambers: the upper vapor chamber **905** and a lower collection chamber **906**. The vapor chamber **905** is defined between the contiguous wall **902** of the cylinder **901** and the walls **907** of the burner tube **960**. The vapor chamber **905** is disposed above the collection chamber **906** and is separated from the collection chamber **906** by a substrate that allows water and vapor to readily pass therethrough, for example, a grating **908**.
(119) The burner **950** is disposed proximal to the top wall **903** of the cylinder **901**. The exemplary burner **950** may be capable of being fueled by any of a variety of types of fuel, such as propane, methane, or natural gas including without limitation flare gas, though some embodiments will use single-fuel burners as will occur to those skilled in the art. The burner tube **960** extends from the burner **950** through the top wall **903** of the cylinder **901**, into and through the vapor chamber **905**, through the grating **908** and may partially extend into collection chamber **906**. Similar to the blower **370**, the blower **970** is configured to propel flame **014** from the burner **950** into and through the burner tube **960** and out into the collection chamber **906** (as is best shown in FIG. **14**).
(120) The force of blower **970** preferably causes the flame **014** to contact and deflect from the bottom wall of collection chamber **906**, and/or from any un-vaporized water that has collected there, and to rise up toward the grating **908**. The collection chamber **906** is configured to serve as an expansion chamber **911** for the flame **014** and the fuel generating the flame **014** fully combusts within the expansion chamber **911** and is converted to heated exhaust gas. This heated exhaust gas rises upwardly through the grating **908** and flows upwardly through the vapor chamber **905**.
(121) To help to prevent deterioration of the burner tube **960**, and to thereby increase the lifespan of the burner tube **960**, the burner tube **960** may be configured in a “floating” or suspended position. For instance, the top end **961** may be configured to securely attach to the top wall **903** while the bottom end **962** may be configured to hang freely. In this configuration, the top end **961** may be securely fastened to the top wall **903** using bolts, screws, welds, clamps, or any other common materials or methods for securing top end **961**.
(122) In the illustrated embodiment, the grating **908** includes a central opening **909** shaped to receive the bottom end **962** of the burner tube **960** therethrough. To guide the burner tube **960** through the central opening **909** while restricting unwanted movement from the burner tube **960**, the central opening **909** may be shaped having a slightly larger diameter in relation to the outer diameter of the burner tube **960** (e.g., but without being too snug as to restrict periodic expanding and contracting of burner tube **960** as temperatures of the materials of the burner tube **960** and grating **908** increase and decrease). As such, it should be understood that the relative sizing of central opening **909**, and in particular the amount of expansion gap **910** in relation to burner tube **960**, can depend on the expansion properties of the materials forming burner tube **960** and expected temperatures of the application.
(123) The vaporization system **900** includes one or more spray nozzles **955**, which perform the same function and replace pipes **401**. The spray nozzles **955** are used to spray and disperse the

wastewater **001**, **003**, **005** into the vapor chamber **905** to be processed. Due to the heavy mineral content of the raw wastewater **001** which can be treated, these spray nozzles **955** may be prone to fouling due to accumulation of mineral build-up, debris, and other contaminants in the raw wastewater **001**.

(124) To prevent the need to fully shut down the vaporization system **900** to clean the spray nozzles **955**, the cylinder **901** of the vaporization system **900** may further include one or more quick-access features **930**. Quick-access features **930** may be positioned to allow access to the vapor chamber **905** and/or the collection chamber **906**. Periodically, the pall rings that are positioned within vapor chamber **905**, such as the stainless-steel pall ring **500** shown in FIG. **5**, require removal for cleaning and/or replacement. The quick-access features **930** may provide a user with an efficient way to access the pall rings **500** to perform those functions.

(125) The quick-access features **930** can each include a door **931**, such as a hinged door pivotable about a hinge **934**, having a securable latch **933** that allows the door **931** to swing open outwardly to provide a user with access to the vapor chamber **905** through a through-hole **932**. Alternatively, the door **931** may be able to disconnect entirely from the quick-access feature **930**, such as by including more than one latch or clamp **933** to secure the door **931** into place at opposing edges (see FIGS. **10**A and **10**B).

(126) The door **931** of the quick-access feature **930** may take the form of a separable plate configured to cover the access opening and be securable thereto by tri-clover clamps **933**, optionally including a gasket not shown positioned between the door **931** and the edges of the access opening to ensure the quick-access features **930** remain leak-free. As shown, one or more quick-access doors **930** may also be positioned on the cylinder **901** adjacent the collection chamber **906** to provide functions similar to those provided by the cleaning port **315** of the vaporization unit **300**.

(127) FIGS. **10**A-B illustrate a collection chamber **990** of an alternative embodiment. The collection chamber **990** may form the lower portion of the vaporization system **900**, in place of collection chamber **906**. The collection chamber **990** is shown with the vapor chamber **905** and the burner tube **960** each removed for clarity. The collection chamber **990** includes an elbow pipe **1002** having a first end **1003** defining a fluid exit opening **1012**, which is fluidly coupled with exterior piping and configured to function as a fluid outlet similar to the fluid outlet **104**, and an open second end **1004** for sucking or otherwise drawing fluid from the lower portion of the collection chamber **990**.

(128) In some applications, sediment **1005** may collect on the floor **1001** of the collection chamber **990** during operation. As such, the elbow pipe **1002**, which is configured with an “L” or elbow shape, may be rotated about an axis **1006** between first and second rotational positions (as shown in the progression from FIG. **10**A to FIG. **10**B) defined centrally relative to the fluid outlet opening on first end **1003** to reduce or prohibit sediment from being sucked or otherwise drawn from floor **1001** of the collection chamber **990** and expelled through the opening **1012** via elbow pipe **1002**. To rotate the elbow pipe **1002** about the axis **1006**, an adjustment rod **1008** may be included with a handle **1009** at one end and a connection feature **1010** at the opposing end. The connection feature **1010** allows the adjustment rod **1008** to be selectively coupled with the elbow pipe **1002** so a user may adjust the angle of the second end **1004** of the elbow pipe **1002** as desired.

(129) To further reduce or prevent sediment **1005** from being expelled from the collection chamber **990** through the elbow pipe **1002**, a solids dike **1014** may be included at or near the second end **1004** to block solids from flowing into the second end **1004** while allowing liquids to freely pass.

(130) In some circumstances during operation of the collection chamber **990**, fluid may develop a circular swirl relative to the floor **1001** within the collection chamber **990** as un-vaporized wastewater **003** is dispensed through the elbow pipe **1002**. To reduce or prevent this fluid swirl, the solids dike **1014** may be further configured to function as a baffle, extending vertically upward from floor **1001**, to interrupt the fluid swirl. It should be understood that the dike and baffle **1014**

may be formed in a "V" shape as shown to assist with the blocking of solids, or it may be formed in any other shape or have an alternative height from the floor **1001** that would be effective for reducing the fluid swirl and liquid movement within the collection chamber **990** while also preventing solids from entering the elbow pipe **1002**.

(131) Referring now to FIGS. **11-13**, the spray rail assemblies **955** for use with vaporization system **900** will now be described. One or more of the spray rail assemblies **955** may be inserted into the vaporization chamber **905** for introducing wastewater **001**, **003**, **005** therein. The spray rail assemblies **955** are configured so that they can be easily removed and maintained or replaced, as needed. With reference to FIG. **11**, the spray rail assembly **955** comprises a rail pipe **980** with a first end **981** and a second end **982**. The spray rail assembly **955** may be hollow to therefore permit wastewater **001**, **003**, **005** to be transferred through the spray rail assembly **955**. Wastewater **001**, **003**, **005** enters the spray rail assembly **955** at the first end **980** through the water inlet **983**. As the wastewater **001**, **003**, **005** can be pressurized, it is pushed through the rail pipe **981** and through the spray rail assembly **955** to exit the spray rail assembly **955** through the plurality of spray nozzles **986**. The second rail end **982** may be a solid cap to prevent the wastewater **001**, **003**, **005** from exiting the spray rail assembly **955** through the second end **982**, or it may incorporate another nozzle **986** at its second end **982**.

(132) In some embodiments, the rail pipe **980** of the spray rail assembly **955** includes a curved body to allow for a better fit within the vapor chamber **905**. In one example, three spray rail assemblies **955** may be utilized within the vapor chamber **905**, with each spray rail assembly **955** configured to curve approximately ⅓ of the distance around the burner tube **960**. The curved body of the rail pipe **980** further ensures wastewater **001**, **003**, **005** expelled from the spray rail assembly **955** is expelled from a central position between the burner tube **960** and the cylinder **901**. The curved body of the rail pipe **980** also ensures that one may efficiently insert and remove the spray rail assembly **955** into and from the vapor chamber **905** through its respective access opening. A flange assembly **984** provides a mounting structure for attaching the spray rail assembly **955** to an exterior surface of the cylinder **901**. Particularly, the flange assembly **984** is affixed or otherwise secured with the spray rail assembly **955** near the first end **981** of the spray rail assembly **955**.

(133) The rail pipe **980** of the spray rail assembly **955** includes a plurality of spray nozzles **986** that are dispersed along the length of rail pipe **980**. In one embodiment, the spray nozzles **986** are holes drilled through the surface of rail pipe **980** to allow wastewater **001**, **003**, **005** from inside rail pipe **980** to spray into vapor chamber **905** via spray nozzles **986** as wastewater droplets **020**. It should be understood that, in alternative versions, it may be desired to use replaceable nozzle assemblies or other openings in lieu of drilled holes in rail pipe **980**.

(134) As illustrated in FIG. **13**, a spray rail wastewater supply line **992** enters the spray rail receiving assembly **988** and is affixed to the assembly **988** by a spray rail assembly clamp **991**. The spray rail wastewater supply line **992** further comprises a spray rail water supply shutoff valve **994**. In some embodiments, each of the plurality of spray rail assemblies **955** would have its own individual water supply shutoff valve **994**. The water supply shutoff valve(s) **994** may be a manually actuated valve, electronically actuated valve, or use alternative or additional. The spray rail assembly clamp **991** may be, for example, a tri-clover type of clamp assembly, or it may be any other type of clamping connector as would occur to one knowledgeable in the art.

(135) FIG. **14** depicts an enlarged view of the area depicted as **913** in FIG. **9**. FIG. **14** depicts the collection chamber **906** of the vaporization system **900** as during use. As is illustrated, the flame **014** is expelled from a discharge **915** of the burner tube **960**. The collection chamber **906** is configured to serve as an expansion chamber **911**. The flame **014** is expelled from the discharge **915** and preferably contacts the un-vaporized wastewater **003**. The flame **014** is then directed outwardly to rise upwardly toward the grating **908**. Falling wastewater droplets **020** will fall through an interact with the flame **014**.

(136) The volume provided by the expansion chamber **911** is preferably sufficient for the fuel

generating the flame **014** to fully combust within the expansion chamber **911** (e.g., converting the fuel and flame **014** to heated exhaust gases). These heated exhaust gases then flow upwardly through the grating **908** and into the vapor chamber **905**.
(137) While the invention has been described in connection with what is presently considered to be the most practical and preferred embodiment, it is to be understood that the invention is not to be limited to the disclosed embodiments, but on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as permitted under the law. Accordingly, the scope of the present invention should be considered in terms of the following claims and is understood not to be limited to the details of structure and operation shown and described in the specification and drawings.
(138) It should be understood that while the use of the word preferable, preferably, or preferred in the description above indicates that feature so described may be more desirable, it nonetheless may not be necessary and any embodiment lacking the same may be contemplated as within the scope of the invention, that scope being defined by the claims that follow.
(139) In reading the claims it is intended that when words such as “a,” “an,” “at least one” and “at least a portion” are used, there is no intention to limit the claim to only one item unless specifically stated to the contrary in the claim. Unless specifically stated to the contrary in the claim, the language “at least one of X, Y, and Z” should be interpreted as including both the conjunctive and disjunctive forms. Specifically, the language “at least one of X, Y, and Z” is intended to encompass the following permutations of X, Y, and Z: X alone; Y alone; Z alone; X and Y; X and Z; Y and Z; and X, Y, and Z. Further, when the language “at least a portion” and/or “a portion” is used the item may include a portion and/or the entire item unless specifically stated to the contrary.

Claims

1. An apparatus, comprising: a chamber having an outer wall; a wastewater introduction port located at an upper portion of the chamber, wherein the wastewater introduction port is configured to distribute wastewater throughout the upper portion; a burner in flow communication with the chamber, wherein the burner is configured to generate flame to be introduced into a lower portion of the chamber through a burner tube, wherein the flame heats an outer wall of the burner tube; and wherein a portion of the wastewater is heated by the outer wall of the burner tube and vaporized in the upper portion of the chamber.
2. The apparatus of claim 1, wherein the flame has a temperature in the range of 500° F. to 3500° F. when introduced into the lower portion.
3. The apparatus of claim 2, wherein the flame interacts with at least one volatile organic compound in the chamber to ignite and incinerate the at least one volatile organic compound.
4. The apparatus of claim 2, further comprising a permeable substrate located between the upper portion and the lower portion, wherein the upper portion is an upper region, wherein the lower portion is a lower region, wherein a vaporization medium is located in the upper region, and wherein the wastewater introduction port is configured to expel the wastewater toward the vaporization medium.
5. The apparatus of claim 4, further comprising a vent in flow communication with the upper region of the chamber, wherein the portion of vaporized wastewater is vented to the atmosphere via the vent, and wherein the chamber remains at substantially atmospheric pressure.
6. The apparatus of claim 5, further comprising: a burner tube extending from the burner through the upper region and toward the lower region, wherein the upper region is defined between the outer wall of the chamber and an outer wall of the burner tube; and wherein the lower region of the chamber is configured to collect un-vaporized wastewater, wherein the burner tube is configured to

direct the flame downwardly at a surface of the collected un-vaporized wastewater, and wherein the flame contacts the surface of the collected un-vaporized wastewater.

7. The apparatus of claim 1, further comprising a vaporization medium located in the upper portion of the chamber, wherein the flame is introduced into the lower portion of the chamber, wherein the lower portion is configured to serve as an expansion chamber for the flame, wherein the fuel generating the flame is fully combusted and converted into heated exhaust gas in the lower portion, and wherein the heated exhaust gas is directed upwardly toward the vaporization medium and flows through the vaporization medium.

8. The apparatus of claim 7, wherein a substantial portion of the wastewater is vaporized in the upper portion, wherein un-vaporized droplets of the wastewater move downwardly from the upper portion into the lower portion, and wherein the flame contacts the un-vaporized droplets as they move downwardly in the lower portion.

9. The apparatus of claim 1, wherein the burner tube is configured to direct the flame toward an un-vaporized portion of the wastewater, wherein the flame contacts an upper surface of the un-vaporized portion.

10. The apparatus of claim 2, wherein the burner tube is configured to direct the flame to contact un-vaporized droplets of the wastewater as the un-vaporized droplets move downwardly through the lower portion of the chamber.

11. The apparatus of claim 2 wherein the lower portion of the chamber is an expansion chamber for the flame.

12. The apparatus of claim 4 wherein the vaporization medium comprises a plurality of pall rings.